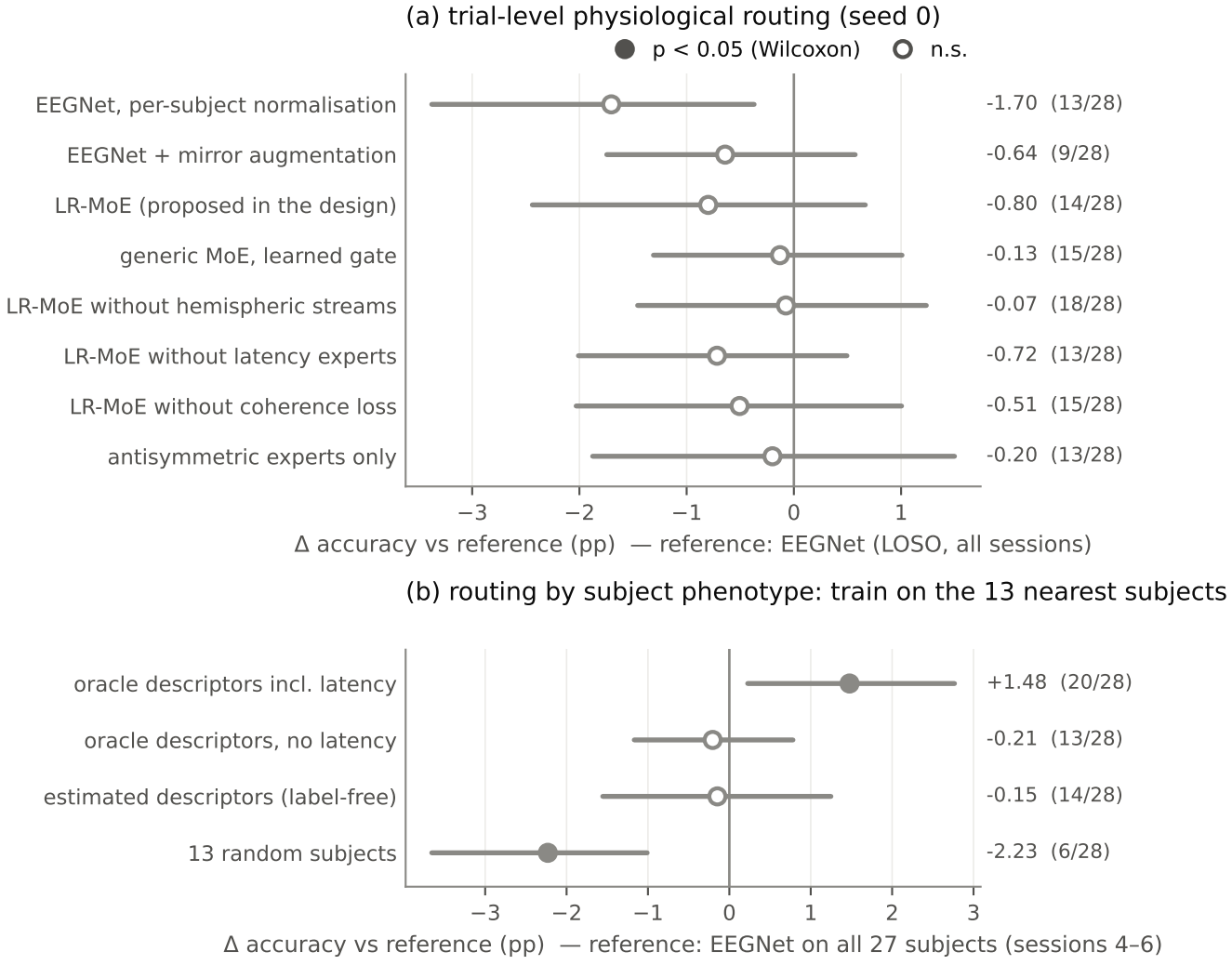
(a) trial-level physiological routing (seed 0)
p < 0.05 (Wilcoxon)
n.s.
EEGNet, per-subject normalisation
EEGNet + mirror augmentation
LR-MoE (proposed in the design)
generic MoE, learned gate
LR-MoE without hemispheric streams
LR-MoE without latency experts
LR-MoE without coherence loss
antisymmetric experts only
-1.70 (13/28)
-0.64 (9/28)
-0.80 (14/28)
-0.13 (15/28)
-0.07 (18/28)
-0.72 (13/28)
-0.51 (15/28)
-0.20 (13/28)
−3
−2
−1
0
1
Δ accuracy vs reference (pp) — reference: EEGNet (LOSO, all sessions)
(b) routing by subject phenotype: train on the 13 nearest subjects
oracle descriptors incl. latency
oracle descriptors, no latency
estimated descriptors (label-free)
13 random subjects
+1.48 (20/28)
-0.21 (13/28)
-0.15 (14/28)
-2.23 (6/28)
−3
−2
−1
0
1
2
3
Δ accuracy vs reference (pp) — reference: EEGNet on all 27 subjects (sessions 4–6)